SAVEETHA

INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Programme: **B.E / B.Tech**

Course Code: **MLA02**

Course Name: **FUNDAMENTALS OF MACHINE LEARNING**

Duration: **1:30 Hr**

Date : 5/11/26

Max Marks: **50 Marks**

ASSIGNMENT – II

UNIT II – SUPERVISED LEARNING

Topics: *Linear regression and classification, naïve Bayes, discriminant functions, logistic regression, decision trees and pruning, neural networks and back-propagation, SVM, ensembles (bagging, boosting)*

Course Outcome

On successful completion of this assignment, students will be able to:

CO	Course Outcome Statement
CO2	Apply supervised learning algorithms for regression, classification, and predictive modeling tasks to solve structured data problems in practical applications.

Answer all the Tasks

Task 1. The following data relate the number of training epochs x to a validation score y.

x	1	2	3	4	5
y	2.1	3.9	6.2	7.8	10.1

(a) Fit $y = w_0 + w_1x$ by ordinary least squares using the normal equations. (b) Predict y at x = 6. (c) Compute the residual sum of squares, MSE and R^2. (d) Analyse the reliability of predicting at x = 6.

(5) (CO2) (BL4)

Task 2. For mean-centred data with $Sxx = \Sigma x^2 = 10$ and $Sxy = \Sigma xy = 19.9$ (single feature, no intercept needed), compute the ridge-regression slope for $\lambda = 5$. Calculate the percentage shrinkage relative to the least-squares slope and explain the bias–variance effect of increasing λ.

(2) (CO2) (BL4)

Task 3. An e-mail filter uses two binary features: Offer (contains the word 'offer') and Link (contains a hyperlink). Ten labelled e-mails are available.

#	Offer	Link	Class
1	Y	Y	Spam
2	Y	N	Spam
3	Y	Y	Spam
4	N	Y	Spam
5	Y	Y	Spam

#	Offer	Link	Class
6	N	N	Ham
7	N	Y	Ham
8	Y	N	Ham
9	N	N	Ham
10	N	N	Ham

(a) Estimate the priors and the class-conditional probabilities without smoothing. (b) Classify an e-mail with Offer = Y, Link = Y and report the posterior P(Spam | x). (c) Repeat (b) with Laplace smoothing ($\alpha = 1$). (d) Classify an e-mail with Offer = N, Link = N (with smoothing) and analyse the effect of the conditional-independence assumption and of smoothing.

(6) (CO2) (BL4)

Task 4. A logistic-regression classifier $p = \sigma(w_1x_1 + w_2x_2 + b)$ is initialised with $w = (0, 0)$, $b = 0$ and trained on two samples: $x^{(1)} = (1, 2)$ with $y = 1$, and $x^{(2)} = (2, 1)$ with $y = 0$. Perform one batch gradient-descent step with learning rate $\eta = 0.5$ on the mean cross-entropy loss. Report the gradient, the updated parameters, the predicted probabilities and the loss before and after the update, and analyse the result.

(4) (CO2) (BL4)

Task 5. A node contains 10 training samples (6 Yes, 4 No). A candidate attribute A splits it into $A = a_1$ (4 samples: 4 Yes, 0 No) and $A = a_2$ (6 samples: 2 Yes, 4 No). Calculate the entropy of the node, the information gain of the split and the Gini-impurity decrease, and evaluate whether the split is worthwhile.

(3) (CO2) (BL4)

Task 6. In a decision tree trained on 100 samples, an internal node t with a four-leaf subtree T_t makes 3 training errors, whereas collapsing t into a single leaf would make 8 errors. Using cost-complexity pruning with $R_\alpha(T) = R(T) + \alpha|\text{leaves}|$, compute the critical value α at which the subtree and the single leaf have equal cost, and evaluate whether the subtree should be pruned for $\alpha = 0.01$ and $\alpha = 0.02$.

(3) (CO2) (BL5)

Task 7. A network has two inputs, one sigmoid hidden unit and one sigmoid output unit (no biases): $h = \sigma(w_1x_1 + w_2x_2)$, $\hat{y} = \sigma(vh)$, with $w_1 = 0.4$, $w_2 = 0.3$, $v = 0.5$. For the input $x = (1, 2)$ with target $t = 1$ and squared-error loss $E = \frac{1}{2}(t - \hat{y})^2$, perform one forward pass and one back-propagation update with $\eta = 0.5$. Report h, $\hat{y}$, E, the error terms δ, the gradients, the updated weights and the new loss.

(4) (CO2) (BL4)

Task 8. A linear SVM in two dimensions has $w = (0.5, 0.5)$ and $b = -2$, with the support vectors (3, 3) (class +1) and (1, 1) (class −1). (a) Verify that the support vectors lie on the margin hyperplanes and compute the geometric margin width. (b) Classify the points (4, 2) and (1, 2) and compute their distances from the decision boundary. (c) Compute the hinge loss for (1, 2) with true label −1 and for (2, 2) with true label +1, and state which points violate the margin.

(4) (CO2) (BL4)

Task 9. Independent classifiers, each with accuracy 0.70, are combined by majority voting. Calculate the ensemble accuracy for 3 and 5 classifiers, comment on the gain from adding more classifiers, and analyse what happens to this gain if the classifiers' errors are strongly correlated.

(3) (CO2) (BL4)

Fundamentals of Machine Learning

Assignment - II

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Course Code: MLA0204

Date: 31/10/26

1.) Given:

x	1	2	3	4	5
y	2.1	3.9	6.2	7.8	10.1

a.) $y = w_0 + w_1 x$

$$\begin{bmatrix} n & \Sigma x \\ \Sigma x & \Sigma x^2 \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \end{bmatrix} = \begin{bmatrix} \Sigma y \\ \Sigma xy \end{bmatrix}$$

$n = 5$, $\Sigma x = 15$

$\Sigma x^2 = 5$, $\Sigma y = 30.1$

$\Sigma xy = 110.3$

$$\begin{bmatrix} 5 & 15 \\ 15 & 55 \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \end{bmatrix} = \begin{bmatrix} 30.1 \\ 110.3 \end{bmatrix}$$

$5 w_0 + 15 w_1 = 30.1$

$15 w_0 + 55 w_1 = 110.3$

$w_0 = 0.05$, $w_1 = 1.99$

$\hat{y} = 0.05 + 1.99 x$

b)

$$\hat{y} = 0.05 + 1.99(6)$$

$$\hat{y} = 11.99$$

c) Calculate RSS, MSE & R^2

x	Actual y	Predicted y
1	2.1	2.04
2	3.9	4.03
3	6.2	6.02
4	7.8	8.01
5	10.1	10.00

$$RSS = \Sigma (y_i - \hat{y}_i)^2 = 0.107$$

$$MSE = RSS/n = 0.0214$$

$$R^2 = 1 - \frac{RSS}{TSS}, \quad TSS = 39.748$$

$$R^2 = 1 - \frac{0.107}{39.748} = 0.9973$$

2) Given:

$S_{xx} = 10$, $S_{xy} = 19.9$, & $\lambda = 5$

a) Ridge - Regression slope.

$$w_{OLS} = \frac{S_{xy}}{S_{xx}} = \frac{19.9}{10} = 1.99.$$

$$w_{ridge} = \frac{S_{xy}}{S_{xx} + \lambda} = \frac{19.9}{10+5} = 1.3267.$$

For Ham:

$P(O = Y \mid Ham) = 1/5 = 0.2$

$P(L = Y \mid Ham) = 1/5 = 0.2$

$P(O = N \mid Ham) = 4/5 = 0.8$

$P(L = N \mid Ham) = 4/5 = 0.8$

D) Classify offer = Y, link = Y

$$P(O, L \mid C) = P(O \mid C)\, P(L \mid C)$$

(2)

$$P(SPAM \mid O, L) \propto 0.5\,(0.8)(0.8)$$

$$= 0.32$$

$$P(Ham \mid O, L) \propto 0.5\,(0.2)(0.2)$$

$$= 0.02$$

$$P(SPAM \mid O, L) = 0.9412 = 94.12\%$$

C.) using Laplace

$$P(x_i \mid C) = \frac{N_i + \alpha}{N_c + \alpha K}$$

$$P(O = Y \mid SPAM) = \frac{4+1}{5+2} = 5/7$$

$$P(L = Y \mid SPAM) = 5/7$$

b.) Calculate Percentage shrinkage.

$$\text{Shrinkage} = \frac{W_{OLS} - W_{ridge}}{W_{OLS}} \times 100$$

$$= \frac{1.99 - 1.3267}{1.99} \times 100$$

$$= 33.33\%$$

3.) Given:

Spam emails = 5

Ham emails = 5

Class	Offer = Y	Offer = N	Link = Y	Link = N
Spam	4	1	4	1
Ham	1	4	1	4

a.) Calculate priors and conditional probabilities

(3)

$P(\text{Spam}) = \frac{5}{10} = 0.5$

$P(\text{Ham}) = \frac{5}{10} = 0.5$

For SPAM:

$P(O = Y \mid SPAM) = \frac{4}{5} = 0.8$, $P(L = Y \mid Spam) = \frac{4}{5} = 0.8$

$P(O = N \mid SPAM) = \frac{1}{5} = 0.2$, $P(L = N \mid SPAM) = \frac{1}{5} = 0.2$

$P(L=Y \mid Ham) = 2/4$

$P(Spam \mid Q, L) = 0.8621 = 86.21\%$

4.) Given:

$P = \sigma(w_1 x_1 + w_2 x_2 + b$

$\sigma(z) = \frac{1}{1 + e^{-z}}$

$w_1 = w_2 = 0, \; b = 0$

(3)

$\eta = 0.5$

$x^{(1)} = (1,2) \quad , \quad y^{(1)} = 1$

$x^{(2)} = (2,1) \quad , \quad y^{(2)} = 0$

a.) $z = w_1 x_1 + w_2 x_2 + b = 0$

$P_1 = P_2 = \sigma(0) = 0.5$

b) Gradient

$$L = -\frac{1}{n} \sum_{i=1}^{n} [y_i \ln P_i + (1-y_i) \ln(1-P_i)]$$

$$\frac{\partial L}{\partial w_j} = \frac{1}{n} \sum_{i=1}^{n} (P_i - y_i) x_{ij}$$

$$\frac{\partial L}{\partial b} = \frac{1}{n} \sum_{i=1}^{n} (P_i - y_i)$$

w_1:

$$\frac{\partial L}{\partial w_1} = \frac{1}{2} [(-0.5)(1) + (0.5)(2)]$$

$$= 0.25$$

w_2:

$$\frac{\partial L}{\partial w_2} = \frac{1}{2}\left[(-0.5)(2) + (0.5)(1)\right]$$

$$= -0.25$$

$$\frac{\partial L}{\partial b} = \frac{1}{2}(-0.5 + 0.5)$$

$$= 0$$

c.) Update the Param:

$$w_j^{new} = w_j^{old} - \eta \frac{\partial L}{\partial w_j}$$

$$w_1 = 0 - 0.5(0.25) = -0.125$$

$$= 0 - 0.5(-0.25) = 0.125$$

$$b = 0 - 0.5(0) = 0$$

d.) updated predictions

$$z_1 = -0.125(1) + 0.125(2) = 0.125$$

$$P_1 = \sigma(0.125)$$

$$P_1 = 0.5312$$

$$z_2 = -0.125(2) + 0.125(1) = -0.125$$

$$P_2 = 0.4688$$

$$\text{Initial} = -\ln(0.5)$$

$$= 0.6931$$

$$L_{new} = 0.6326$$

Given:

Total Samples = 10

Yes = 6

No = 4

Split:

A = a_1 4 yes, 0 No

A = a_2 2 yes, 4 No.

a.) entropy

$$H(S) = -\sum_i P_i \log_2 P_i$$

$$= -\left[\frac{6}{10}\log_2 \frac{6}{10} + \frac{4}{10}\log_2 \frac{4}{10}\right]$$

$$H(S) = 0.9710 \text{ bits}$$

b.) Calculate info gain.

$$H(a_1) = -[1 \log_2 1 + 0] = 0$$

$$H(a_2) = -\left[\frac{2}{6}\log_2 \frac{2}{6} + \frac{4}{6}\log_2 \frac{4}{6}\right]$$

$$H(a_2) = 0.9183 \text{ bits}$$

$$H(S|A) = \frac{4}{10}(0) + \frac{6}{10}(0.9183)$$

$$= 0.5510$$

$$IG = H(S) - H(S|A)$$

$$= 0.9710 - 0.5510$$

$$= 0.4200 \text{ bits}.$$

c.) Gini impurity

$$G(S) = 1 - \sum_i P_i^2$$

$$G(S) = 1 - (0.6^2 + 0.4^2)$$

$$= 0.48.$$

$$G(a_1) = 1 - (1^2 + 0^2) = 0$$

$$G(a_2) = 1 - \left[\left(\tfrac{1}{3}\right)^2 + \left(\tfrac{2}{3}\right)^2\right]$$

$$= \tfrac{4}{9} = 0.444$$

$$G_{after} = \tfrac{4}{10}(0) + \tfrac{6}{10}(0.444)$$

$$= 0.266$$

$$\Delta G = 0.48 - 0.266$$

$$= 0.2133$$

b.) Given:

Total training samples = 100

Errors using four-leaf subtree = 3

Errors after collapsing into one leaf = 8.

$$R_\alpha(t) = R(T) + \alpha(\text{leaves})$$

a.) Calculate critical value of α.

$$R(T_t) = \tfrac{3}{100} = 0.03$$

$$R(T) = \tfrac{8}{100} = 0.08$$

$$0.03 + 4\alpha = 0.08 + \alpha$$

$$3\alpha + = 0.05 \;\therefore\; \alpha_{critical} = 0.0166$$

Pruning for $\alpha = 0.01$

$$R_\alpha(T_t) = 0.03 + 4(0.01)$$

$$= 0.07$$

$$R_\alpha(t) = 0.08 + 0.01$$

$$= 0.09$$

3

$0.07 < 0.09$

Retain

Pruning for $\alpha = 0.02$

$$R_\alpha(T_t) = 0.03 + 4(0.02)$$

$$= 0.11$$

$$R_\alpha(t) = 0.10$$

$0.10 < 0.11$

Prune.

7) Given:

$$h = \sigma(w_1 x_1 + w_2 x_2)$$

$$\hat{y} = \sigma(vh)$$

$w_1 = 0.4$, $w_2 = 0.3$, $v = 0.5$

$x = (1, 2)$, $t = 1$

$$\eta = 0.5$$

$$E = \frac{1}{2}(t - \hat{y})^2$$

a.) Forward propogation :

$$h = \sigma(0.4(1) + 0.3(2))$$

$$= \sigma(1) = 0.7311$$

$$\hat{y} = \sigma(0.5(0.7311))$$

$$= 0.5904$$

$$E = \frac{1}{2}(1 - 0.5904)^2 = 0.0834.$$

b.) Back propogation error terms :-

$$\delta_o = (t - \hat{y})\hat{y}(1 - \hat{y})$$

$$= (1 - 0.5904)(0.5904)(1 - 0.5904)$$

$$= 0.09906.$$

$$\delta_h = h(1-h)v\delta_o$$

$$= 0.009738.$$

c.) Calculate gradients.

$$\frac{\partial E}{\partial v} = -\delta_o h$$

$$\frac{\delta E}{\delta v} = -0.01242.$$

$$\frac{\delta E}{\delta w_1} = -\delta h x_1$$

$$= -0.009738$$

$$\frac{\delta E}{\delta w_2} = -0.01948.$$

d.) update weights:

$$v = 0.53621, \quad w_1 = 0.40487, \quad w_2 = 0.30974$$

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ANSWER SCRIPT EVALUATION SHEET

ASSIGNMENT - II

Programme	B.E / B.Tech	**Course Code**	MLA02
Course Name	**Fundamentals of Machine learning.**	**Date**	5/11/26
Duration	1 Hr 30 Min	**Max Marks**	50

Register No.: _____

TASK 1	Max Marks	Marks Awarded
Task 1	5	4
Task 2	2	2
Task 3	6	3
Task 4	4	3
Task 5	3	3
Task 6	3	3
Task 7	4	2
Task 8	4	–
Task 9	3	–
Task 10	4	–
Task 11	4	–
Task 12	8	–
TOTAL	**50**	20

Total in words: twenty only

20/50

Many question were skipped.